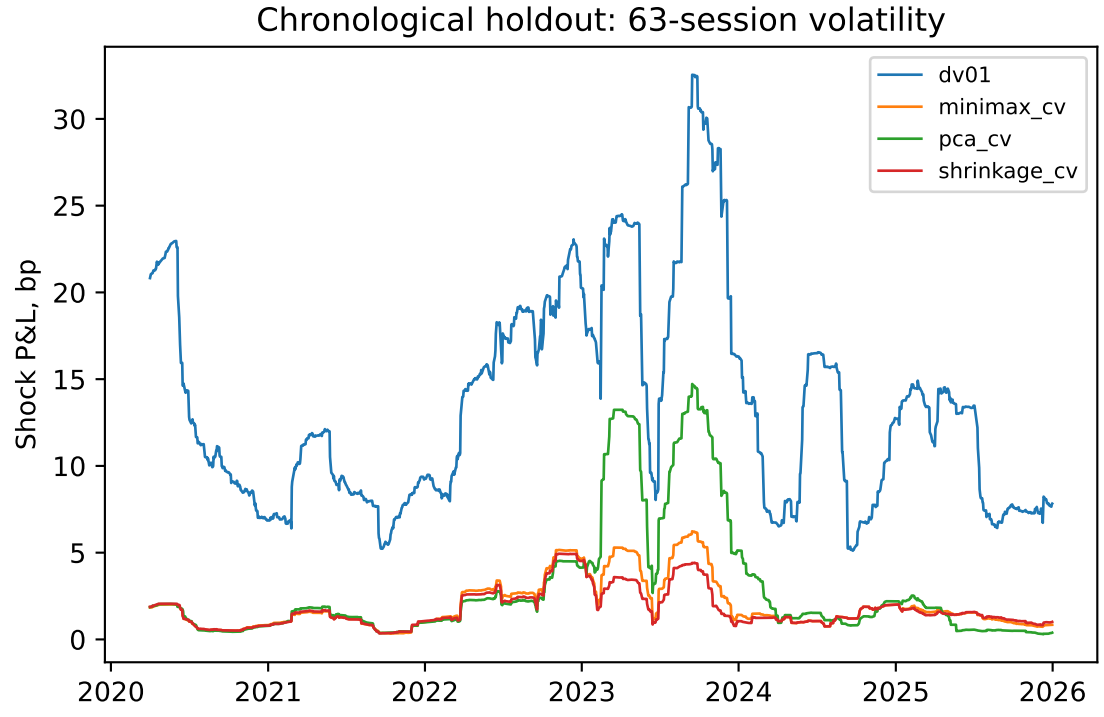
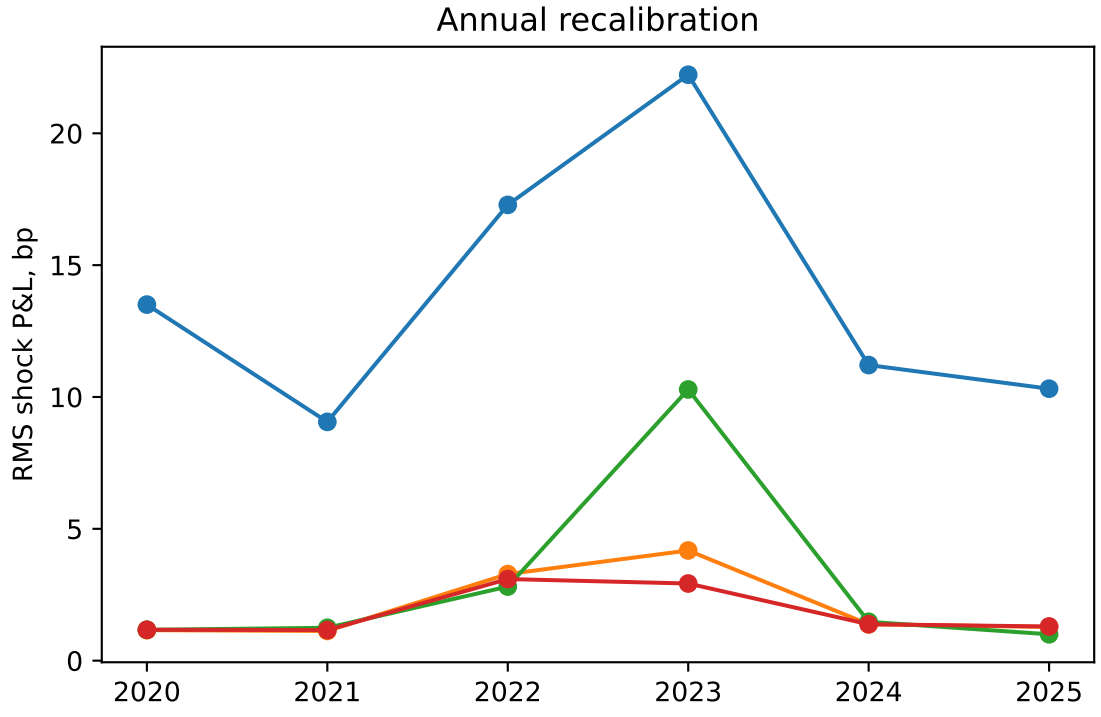
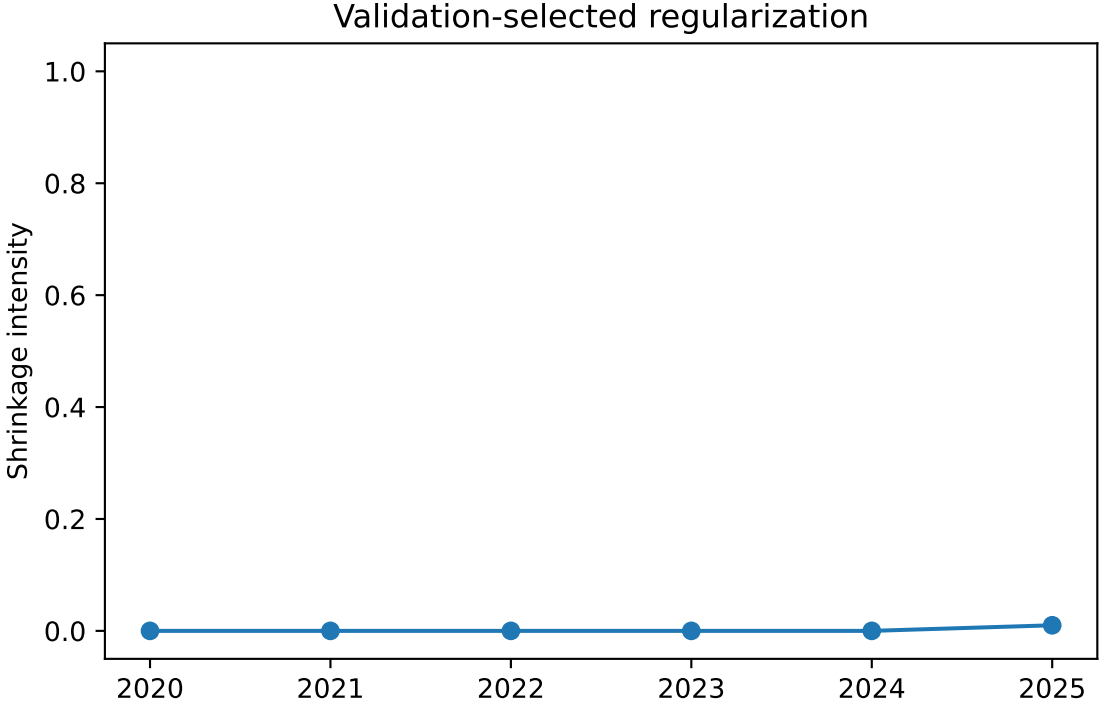
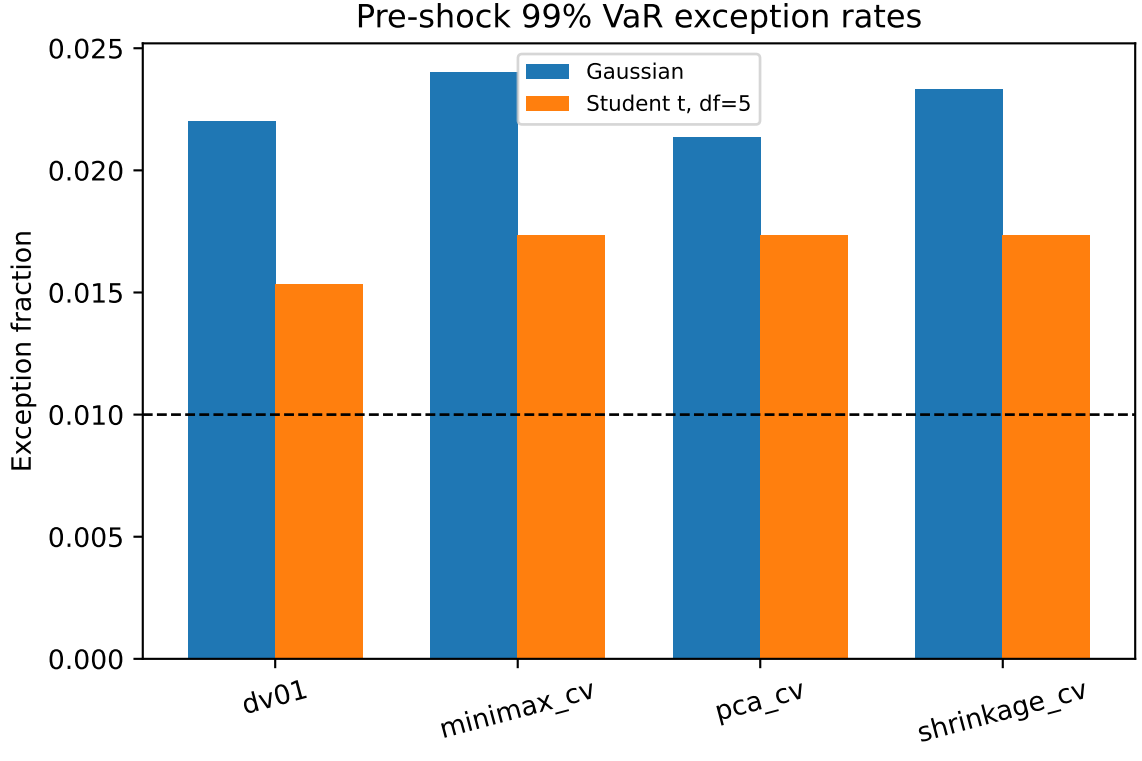

Chronological holdout: 63-session volatility
Shock P&L, bp
dv01
minimax_cv
pca_cv
shrinkage_cv
0
5
10
15
20
25
30
2020
2021
2022
2023
2024
2025
2026


Annual recalibration
RMS shock P&L, bp
0
5
10
15
20
2020
2021
2022
2023
2024
2025


Validation-selected regularization
Shrinkage intensity
0.0
0.2
0.4
0.6
0.8
1.0
2020
2021
2022
2023
2024
2025


Pre-shock 99% VaR exception rates
Exception fraction
Gaussian
Student t, df=5
0.000
0.005
0.010
0.015
0.020
0.025
dv01
minimax_cv
pca_cv
shrinkage_cv